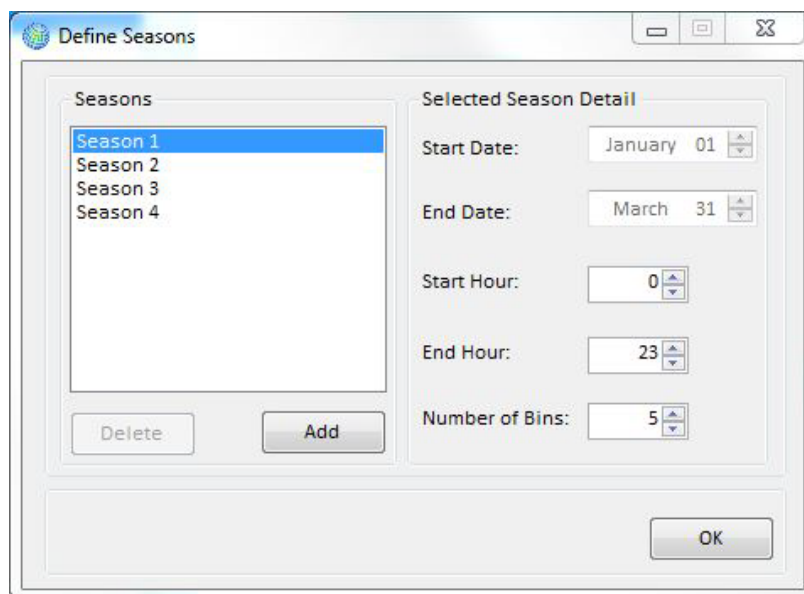


calculate benefits, BenMAP-CE in certain cases needs to generate complete sets of metric values by estimating concentrations for those days that have missing observations. This can be important if certain seasons tend to have more missing values than others.

To define **Seasons** for a **Pollutant**, click the **Define Seasons for all Pollutant Metrics** button. This will bring up the **Define Seasons** window. For each season desired, click the **Add** button, select the appropriate **Start Date** and **End Date**, which define the days included in the season; and the appropriate **Start Hour** and **End Hour**, which define the hours included in monitoring period. The advanced options for PM_{2.5} look like the following:



Once you have finished defining the **Seasons**, click **OK** to return to the **Pollutant Definition** window.

If you later wish to View or Edit a particular **Pollutant** definition, simply select the appropriate **Pollutant** within the **Available Pollutants** box and click the **Edit** button. When you are done, click **OK** to return to the **Modify Datasets** window.

After defining all of the pollutants that you want, click **OK**. This will return you to the **Modify Datasets** window.

4.1.3 Monitor Datasets

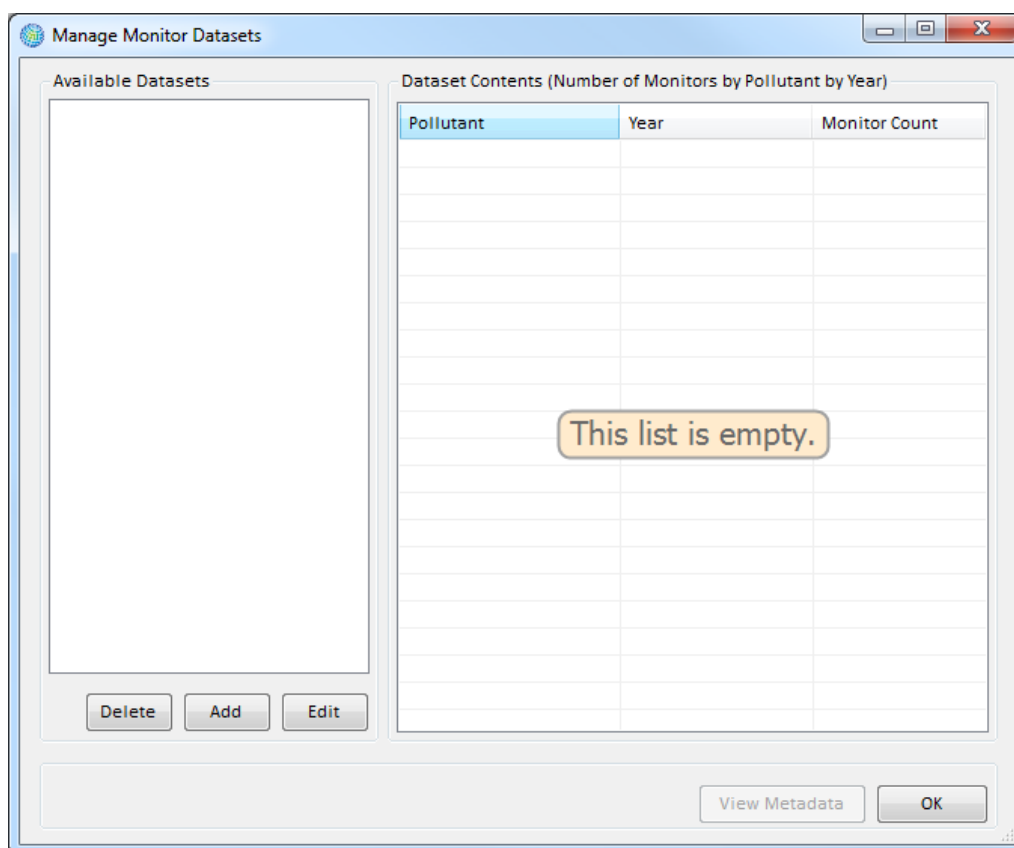
The **Monitor Datasets** section of the **Modify Datasets** window allows you to add air pollution monitoring data to your setup. Air pollution monitoring data may be used to estimate ambient pollution levels in each grid cell defined by a **Grid Definition**.

BenMAP-CE uses a variety of procedures (such as *Voronoi Neighbor Averaging*, discussed later) to interpolate the monitor data points across the area of interest.

NOTE: Air pollution data in BenMAP-CE is of two types: (1) point source monitoring data and (2) **Grid Definition**-based modeling data. Both types of data must be associated with a particular pollutant that you have already defined. Only the point source monitoring data is stored in the setup database. The modeling data are loaded into BenMAP-CE as you need them for a particular analysis.

4.1.3.1 Add Monitor Datasets

To start, click on the **Manage** button below the **Monitor Datasets** box. The **Manage Monitor Datasets** window will appear. From this window you may **Add** monitoring data, view and **Edit** existing datasets, as well as **Delete** them. The section on the left under **Available Datasets** lists the monitor datasets that are currently in the setup. The section on the right under the **Dataset Contents** identifies the number of monitors in each dataset by *Pollutant* and by *Year*. To view the metadata of a particular monitor dataset, select an **Available Dataset** and click on a row from the **Dataset Contents**, then click the **View Metadata** button. This allows you to view further information about references or descriptions of the file.



To start adding data, click the **Add** button. This will bring up the **Monitor Dataset Definition** window. Give the dataset a name in the **Dataset Name** box, choose the

appropriate pollutant from the **Pollutant**¹⁵ drop-down menu, and then type the 4-digit **Year** of the data in the **Year** box.

NOTE: The **Dataset** that you define can have one or more pollutants and multiple years of data (e.g., representing a particular monitoring network). However the data must be imported one pollutant and one year at a time.

Monitor data must be formatted in a database file, with monitor definition information and monitor values in a single line.

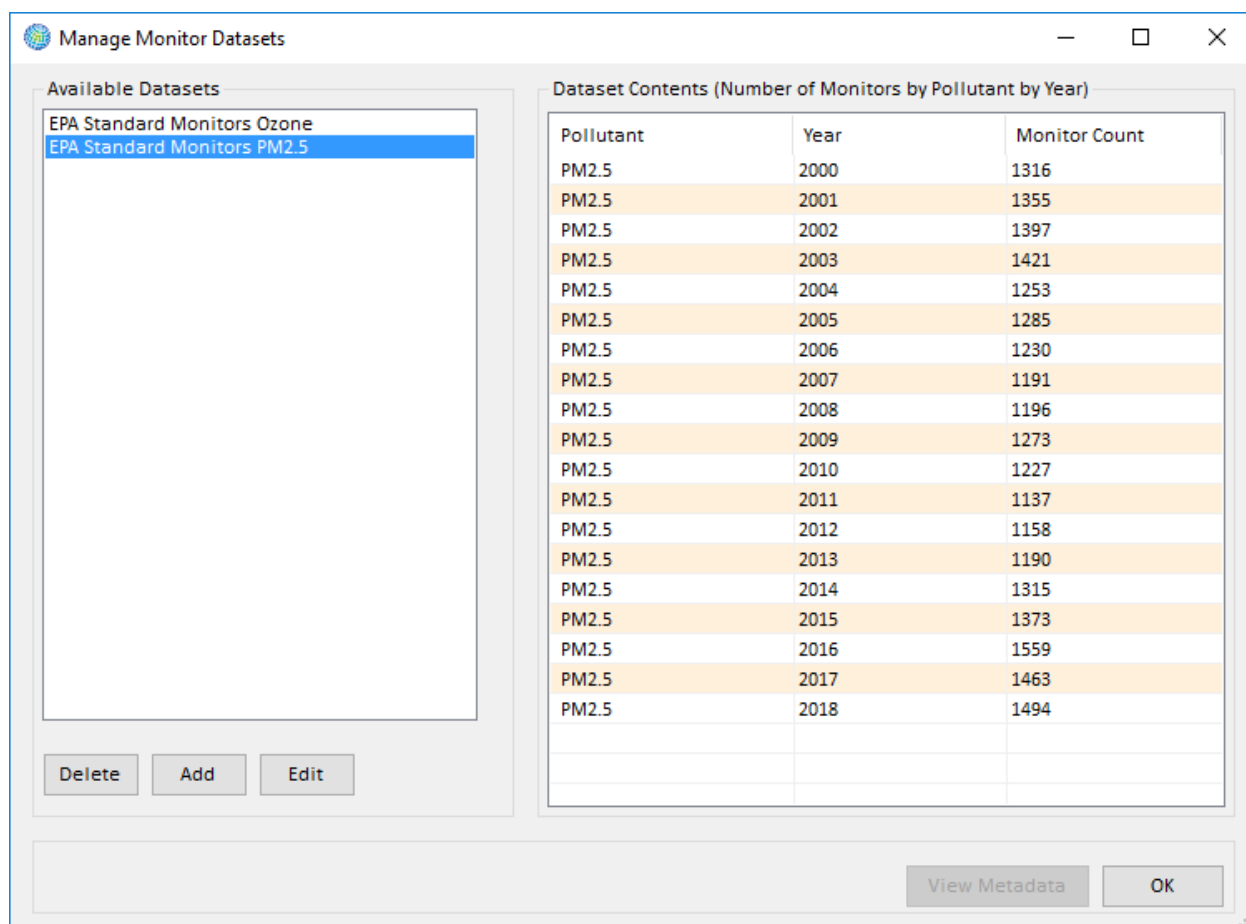
After specifying the **Pollutant** and **Year**, click on the **Load Data From File** button to bring up a window from which you can **Browse** the BenMAP-CE Data directory to find the desired data file. Click **Open**, to choose the file.

Users are advised to click the **Validate** button before loading the monitor dataset. BenMAP-CE's validation tool will review the file format (column names, required

¹⁵ The pollutants in the **Pollutant** drop-down menu have been defined under the **Pollutants** box on the **Modify Datasets** window.

columns, and data types) and provide a report with any errors or warnings.¹⁶ The user also has the option of adding metadata to the file to save references and comments about that specific file. To add metadata, click the **View Metadata** button. After passing validation (and adding metadata if desired) click the **OK** button to bring you back to the **Monitor Dataset Definition** window.

Repeat this procedure to load all of your monitoring data. If you try to load the data for the same pollutant/year combination, BenMAP-CE will warn you of the duplication.



To see the years of data and the number of monitors each year, use the scrollbars on the bottom and on the right of the **Dataset Contents** box. To view the metadata for a particular entry in the dataset, choose an **Available Dataset** and row in the **Dataset Contents** and click the **View Metadata** button.

To delete existing datasets, select the dataset in the **Available Datasets** list and click the **Delete** button. To edit an existing dataset, select the dataset in the **Available Datasets** list and click the **Edit** button. (Note: Certain pre-loaded datasets cannot be

¹⁶ Validation reports are saved to C:\Users\\Documents\My BenMAP-CE Files\ValidationResults.